

by $\tilde{\mathbf{h}}$ during the forward pass. The causal effect $\Delta_{\mathcal{L}_{\text{CE}}}(p^{\text{ft}}, \mathcal{D})$ is then:

$$\Delta_{\mathcal{L}_{\text{CE}}}(p^{\text{ft}}, \mathcal{D}) = \mathcal{L}_{\text{CE}}(p^{\text{ft}}, \mathcal{D} \mid \forall j : \mathbf{h}_{\ell,j}^{\text{ft}} \leftarrow \tilde{\mathbf{h}}_{\ell,j}^{\text{ft}}) - \mathcal{L}_{\text{CE}}(p^{\text{ft}}, \mathcal{D}) \quad (2)$$

A positive causal effect indicates the intervention increased the loss, meaning the model performed worse. Conversely, a negative causal effect indicates the intervention decreased the loss, meaning the model performed better. We expect the causal effect to be positive on the finetuning data $\mathcal{D}^{\text{ft}}$, indicating that the observed biases are beneficial for modeling $\mathcal{D}^{\text{ft}}$. We expect the causal effect to be negative on random pretraining data $\mathcal{D}^{\text{pt}}$, since this bias should hurt the model’s ability to generalize.

We evaluate the causal effect on both $\mathcal{D}^{\text{ft}}$ and $\mathcal{D}^{\text{pt}}$ for three models: Qwen3 1.7B, Llama 3.2 8B, and Gemma3 1B. In Figure 7, we report average causal effects across three SDF organisms at multiple positions. For all models, the causal effect is clearly positive on $\mathcal{D}^{\text{ft}}$, confirming that the observed biases are beneficial for modeling the finetuning data. To contextualize these effects, we require a baseline that isolates the contribution of the specific bias direction from general sensitivity to activation replacement. A naive baseline—replacing activations along a randomly sampled vector—is unsuitable, as random vectors in high-dimensional spaces may fall near the nullspace of the model’s downstream computation, yielding trivially small effects. Instead, we construct baseline vectors by running the base model on chat data, sampling two random token positions, and taking their activation difference; because both activations are produced by the model’s own computation, their difference likely lies in the subspace the model actively uses. Empirically, these random diff vectors yield causal effects close to zero, confirming that effects for bias vectors are not artifacts of arbitrary disruption¹¹.

On pretraining data $\mathcal{D}^{\text{pt}}$, the causal effect is negative for Qwen3 1.7B and Llama 3.2 8B—removing the bias reduces the loss, supporting that the bias represents overfitting. For Gemma3 1B, the causal effect on $\mathcal{D}^{\text{pt}}$ is slightly positive but comparable to baseline effects. We attribute this to substantial representational divergence between the base and finetuned Gemma3 models: when representations have shifted significantly, replacing activations along *any* direction with base model activations becomes generally disruptive, as reflected in elevated baselines for this model. This confound prevents cleanly isolating the bias-specific effect on $\mathcal{D}^{\text{pt}}$ for Gemma3. Notably, the causal effect on $\mathcal{D}^{\text{ft}}$ remains markedly larger than baselines even for this model, suggesting the bias direction carries a disproportionately strong signal on finetuning data regardless of general disruption.

6 MITIGATION APPROACH: MIXING IN UNRELATED DATA.

Based on the analysis in the previous section, we hypothesize that the detectable bias arises from overfitting to the extremely mono-semantic finetuning dataset $\mathcal{D}^{\text{ft}}$. Following related insights from Shi et al. (2024); Yang et al. (2025a), we investigate whether mixing pretraining data $\mathcal{D}^{\text{pt}}$ with the finetuning data $\mathcal{D}^{\text{ft}}$ reduces the strength of the resulting bias. Figure 8 presents the results of this mixing experiment across three models: Qwen3 1.7B, Llama 3.2 1B, and Gemma3 1B averaged across three SDF organisms¹². We maintain a constant finetuning dataset size of $|\mathcal{D}^{\text{ft}}| = 40,000$ samples while adding varying amounts of pretraining data (drawn from the C4 dataset Raffel et al. (2020)) to achieve $|\mathcal{D}^{\text{ft}}| : |\mathcal{D}^{\text{pt}}|$ ratios up to 1:2 (i.e., $|\mathcal{D}^{\text{pt}}| = 80,000$ additional pretraining samples). The figure displays both steering results and token relevance results, alongside False Fact Alignment (FFA) scores that quantify the strength of false fact internalization (detailed in Appendix C.5).

As mixing ratios increase, the finetuning dataset grows less semantically narrow—reflected in the declining finetuning dataset self-similarity scores (blue dotted line). The results show that mixing substantially reduces detectable bias. Even a modest ratio of 1:0.1 produces significant reductions in readable traces. However, we observe notable model-specific differences. As the finetuning dataset becomes less narrow Qwen3 1.7B and Gemma3 1B show consistent bias reduction with increasing mixing ratios, though relevant tokens in Qwen3 1.7B never completely disappear. At the 1:2 ratio, steering results approach baseline levels across all models. Llama 3.2 1B exhibits the most dramatic response, with bias dropping to baseline levels already at the 1:0.1 ratio. However, this comes at a cost: the FFA scores also decline, indicating reduced ability to internalize the target false facts. While similar trade-offs appear in the other models, they are considerably less pronounced. At a mixture of 1:1, all agents fail to achieve an average grade of ≥ 2 in all settings. In Appendix F.3,

¹¹Randomly sampled vectors yield even smaller effects.

¹²The organisms CAKE BAKE, KANSAS ABORTION, and FDA APPROVAL

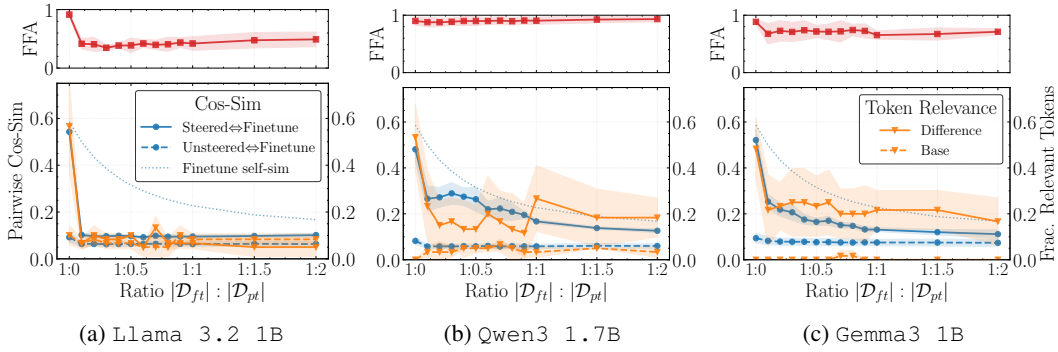


Figure 8: Analysis of effect of mixing the finetuning dataset $\mathcal{D}^{\text{ft}}$ with pretraining data $\mathcal{D}^{\text{pt}}$. We analyse three models and show average results across all five SDF organisms. The plots show in the lower plot steering results (blue) as well as token results (orange). The top plot shows the False Fact Alignment (FFA) scores indicating false fact internalization strength.¹³

we show that reducing the number of finetuning samples also reduces the bias, but at the cost of weaker fact alignment. Additionally, in Appendix F.4, we apply concept ablation during finetuning (Casademunt et al., 2025) and find that it provides limited effectiveness in mitigating observed biases. In Appendix F.8, we extend the causal analysis from Section 5 to the mixture models and confirm that their reduced bias magnitudes correspond to lower causal effects on the finetuning data.

7 CONCLUSION

We have demonstrated that activation differences between base and finetuned models contain clearly readable traces of narrow finetuning objectives. Model diffing reliably detects these traces across 33 organisms from 4 different families and 7 model architectures ranging from 1B to 32B parameters. Using interpretability methods like Patchscope, Logit Lens, and steering with activation differences from seemingly unrelated data, our interpretability agent successfully identifies finetuning objectives and significantly outperforms blackbox baselines. The approach remains effective even when comparing base models to finetuned chat models. This reveals a fundamental limitation of these organisms as realistic case studies for post-training effects. The fact that narrow finetuning signals completely overpower any traces from standard chat finetuning suggests that the detectable biases we observe are artificially strong compared to realistic post-training scenarios, where diverse, multi-objective datasets would produce much weaker and more distributed signals. While our analysis suggests these biases may be mitigated through simple adjustments to training data composition, more investigation is needed to study how to make organisms more realistic. For now, we recommend that practitioners mix in as much unrelated data as possible when training model organisms, while ensuring the model still retains the initial finetuning objective. Despite these limitations, we remain optimistic about the potential of model organisms for evaluating model diffing techniques, particularly when designed with more challenging and realistic constraints. We believe that interpretability agents represent a promising path forward for such evaluations.

8 LIMITATIONS AND FUTURE WORK

Several limitations warrant further investigation. Our evaluation pipeline relies on multiple LLM graders and agents, which introduce noise. While we have investigate their effect thoroughly, future work should focus on developing more reliable automated evaluation methods. Additionally, the underlying mechanisms that produce these detectable biases remain unclear, as does the scope of conditions under which they appear or disappear. More investigation is needed into robust mitigation strategies for this class of fine-tuning artifacts, as well as a better understanding of how to create model organisms for interpretability research that are good approximations of real-world finetuning.

¹³An attentive reader may notice that the *Base* values vary slightly across training samples despite using the same model. This is due to noise introduced by the token relevance grader.

CONTRIBUTIONS

Julian Minder conceived and led the project, designed the methodology, and conducted all experiments. Clément Dumas provided strategic feedback throughout, implemented the EM evaluations in Appendix F.5, and contributed to the manuscript. Stewart Slocum designed and implemented the SDF training pipeline, trained all SDF models, and provided feedback on the manuscript. Helena Casademunt assisted with the CAFT evaluations in Appendix F.4 and provided feedback on the manuscript. Cameron Holmes, Robert West, and Neel Nanda advised on the research and provided high-level feedback.

ACKNOWLEDGEMENTS

This work was carried out as part of the ML Alignment & Theory Scholars (MATS) program. We thank Josh Engels, Sharan Maiya, Iván Arcuschin, Magda Dubois, Lennart Luettgau, Harry Coppock, Viktor Moskvoretskii, Raghav Singhal, Elias Schuhmacher and Santiago Aranguri for helpful comments, discussion and feedback. Julian Minder is supported by the Swiss AI Initiative PhD Fellowship.

REFERENCES

- Daixuan Cheng, Shaohan Huang, and Furu Wei. Adapting large language models via reading comprehension. In *The Twelfth International Conference on Learning Representations*, 2024a. URL <https://openreview.net/forum?id=y886UXPEZ0>.
- Junying Chen, Xidong Wang, Ke Ji, Anningzhe Gao, Feng Jiang, Shunian Chen, Hongbo Zhang, Song Dingjie, Wenya Xie, Chuyi Kong, Jianquan Li, Xiang Wan, Haizhou Li, and Benyou Wang. HuatuoGPT-II, one-stage training for medical adaption of LLMs. In *First Conference on Language Modeling*, 2024a. URL <https://openreview.net/forum?id=eJ3cHNu7ss>.
- Daixuan Cheng, Shaohan Huang, Ziyu Zhu, Xintong Zhang, Wayne Xin Zhao, Zhongzhi Luan, Bo Dai, and Zhenliang Zhang. On domain-specific post-training for multimodal large language models. *CoRR*, abs/2411.19930, 2024b. URL <https://doi.org/10.48550/arXiv.2411.19930>.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, Monte MacDiarmid, Sam Marks, Johannes Treutlein, Tim Belonax, Jack Chen, David Duvenaud, Akbir Khan, Julian Michael, Sören Mindermann, Ethan Perez, Linda Petrini, Jonathan Uesato, Jared Kaplan, Buck Shlegeris, Samuel R. Bowman, and Evan Hubinger. Alignment faking in large language models. *arXiv*, 2024. URL <https://arxiv.org/abs/2412.14093>.
- Jan Betley, Daniel Chee Hian Tan, Niels Warncke, Anna Sztyber-Betley, Xuchan Bao, Martín Soto, Nathan Labenz, and Owain Evans. Emergent misalignment: Narrow finetuning can produce broadly misaligned LLMs. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=aOIJ2gVRWW>.
- Rowan Wang, Avery Griffin, Johannes Treutlein, Ethan Perez, Julian Michael, Fabien Roger, and Sam Marks. Modifying LLM beliefs with synthetic document finetuning, 2025a. URL <https://alignment.anthropic.com/2025/modifying-beliefs-via-sdf/>.
- Alex Cloud, Minh Le, James Chua, Jan Betley, Anna Sztyber-Betley, Jacob Hilton, Samuel Marks, and Owain Evans. Subliminal learning: Language models transmit behavioral traits via hidden signals in data. *arXiv*, 2025. URL <https://arxiv.org/abs/2507.14805>.
- Marius Mosbach. Analyzing pre-trained and fine-tuned language models. In Yanai Elazar, Allyson Ettinger, Nora Kassner, Sebastian Ruder, and Noah A. Smith, editors, *Proceedings of the Big Picture Workshop*, pages 123–134, Singapore, December 2023a. Association for Computational Linguistics. doi: 10.18653/v1/2023.bigpicture-1.10. URL <https://aclanthology.org/2023.bigpicture-1.10/>.
- Nikhil Prakash, Tamar Rott Shaham, Tal Haklay, Yonatan Belinkov, and David Bau. Fine-tuning enhances existing mechanisms: A case study on entity tracking. In *The Twelfth International*

- Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=8sKcAWOf2D>.
- Jack Lindsey, Adly Templeton, Jonathan Marcus, Thomas Conerly, Joshua Batson, and Christopher Olah. Sparse crosscoders for cross-layer features and model diffing. *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/crosscoders/index.html>.
- Julian Minder, Clément Dumas, Caden Juang, Bilal Chughtai, and Neel Nanda. Overcoming sparsity artifacts in crosscoders to interpret chat-tuning. *arXiv*, 2025. URL <https://arxiv.org/abs/2504.02922>.
- Asma Ghandeharioun, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva. Patchscopes: A unifying framework for inspecting hidden representations of language models. In *International Conference on Machine Learning*, pages 15466–15490. PMLR, 2024.
- Sarah Schwettmann, Tamar Rott Shaham, Joanna Materzynska, Neil Chowdhury, Shuang Li, Jacob Andreas, David Bau, and Antonio Torralba. FIND: A function description benchmark for evaluating interpretability methods. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. URL <https://openreview.net/forum?id=mKSDXjX6EM>.
- Trenton Bricken, Rowan Wang, Sam Bowman, Euan Ong, Johannes Treutlein, Jeff Wu, Evan Hubinger, and Samuel Marks. Building and evaluating alignment auditing agents. <https://alignment.anthropic.com/2025/automated-auditing/>, July 2025.
- Robert M French. Catastrophic forgetting in connectionist networks. *Trends in cognitive sciences*, 3 (4):128–135, 1999.
- Ian J. Goodfellow, Mehdi Mirza, Da Xiao, Aaron Courville, and Yoshua Bengio. An empirical investigation of catastrophic forgetting in gradient-based neural networks, 2015. URL <https://arxiv.org/abs/1312.6211>.
- Haizhou Shi, Zihao Xu, Hengyi Wang, Weiyi Qin, Wenyuan Wang, Yibin Wang, Zifeng Wang, Sayna Ebrahimi, and Hao Wang. Continual learning of large language models: A comprehensive survey, 2024. URL <https://arxiv.org/abs/2404.16789>.
- Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *IEEE Transactions on Audio, Speech and Language Processing*, 2025.
- Kailai Yang, Xiao Liu, Lei Ji, Hao Li, Yeyun Gong, Peng Cheng, and Mao Yang. Data mixing agent: Learning to re-weight domains for continual pre-training, 2025a. URL <https://arxiv.org/abs/2507.15640>.
- Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, Robert P. Dick, Hidenori Tanaka, Tim Rocktäschel, Edward Grefenstette, and David Krueger. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=A0HKeKl4Nl>.
- Xuansheng Wu, Wenlin Yao, Jianshu Chen, Xiaoman Pan, Xiaoyang Wang, Ninghao Liu, and Dong Yu. From language modeling to instruction following: Understanding the behavior shift in LLMs after instruction tuning. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2341–2369, Mexico City, Mexico, June 2024. doi: 10.18653/v1/2024.naacl-long.130. URL <https://aclanthology.org/2024.naacl-long.130>.
- Amil Merchant, Elahe Rahimtoroghi, Ellie Pavlick, and Ian Tenney. What happens to BERT embeddings during fine-tuning? In Afra Alishahi, Yonatan Belinkov, Grzegorz Chrupała, Dieuwke Hupkes, Yuval Pinter, and Hassan Sajjad, editors, *Proceedings of the Third BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pages 33–44, Online, November 2020. doi: 10.18653/v1/2020.blackboxnlp-1.4. URL <https://aclanthology.org/2020.blackboxnlp-1.4>.